

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Enhancing Healthcare Performance Through Data - Driven Insights

Project Domain

Power BI with Copilot

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:



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Year - 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my mentor, Umamaheshwari Ma'am, for her constant guidance, encouragement, and valuable suggestions throughout the completion of this project. Her support and motivation helped me understand the concepts of data analytics, visualization, and AI tools more effectively.

I would also like to thank Edunet Foundation, LinkedIn, Microsoft, and SAP for organizing the AICW Fellowship Program and providing an excellent platform to learn practical applications of Artificial Intelligence, Power BI, Excel, and Microsoft Copilot. The learning resources and mentorship provided during the fellowship greatly enhanced my technical knowledge and professional skills.

I am thankful to my institution and faculty members for providing the necessary support and facilities required to successfully complete this project work.

I would also like to extend my heartfelt thanks to my family and friends for their continuous encouragement, motivation, and support throughout the project duration.

Finally, I am grateful for this opportunity, which helped me gain practical exposure to healthcare analytics, business intelligence tools, and AI-assisted reporting techniques.

ABSTRACT OF THE PROJECT

This project focuses on enhancing healthcare performance through data driven insights using Power BI, Excel, and AI tools such as Microsoft Copilot.

The objective of the project is to analyse patient data and identify operational inefficiencies, resource utilisation patterns, and healthcare performance indicators.

The project involves collecting, cleaning, and preprocessing healthcare datasets obtained from open-source platforms such as Kaggle. Various data cleaning techniques were applied to remove duplicate records, handle missing values, and standardize the dataset for accurate analysis.

Interactive dashboards and visualizations were created using Power BI to represent patient demographics, admission statistics, disease trends, and hospital operational performance. AI-powered tools such as Microsoft Copilot were used to generate summaries and insights efficiently.

The project successfully demonstrates how business intelligence and AI tools can transform raw healthcare data into meaningful insights that support better healthcare decision-making.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

Healthcare institutions generate large amounts of patient and operational data every day. However, inconsistent, incomplete, and unreliable data often creates challenges in understanding patient needs, hospital performance, and resource utilisation.

Without proper data analysis, decision-makers face difficulties in managing patient flow and optimising resources. Existing systems often fail to provide actionable insights in a clear and efficient manner.

1.2 Motivation

The motivation behind this project is to understand how data analytics and AI tools can improve healthcare management and decision-making.

Healthcare data contains valuable insights that can help hospitals identify inefficiencies, improve patient care, and optimise resource allocation. This project also aligns with modern industry trends in AI-assisted analytics and business intelligence.

1.3 Objectives

1. To collect and understand healthcare datasets.
2. To clean and preprocess raw healthcare data.
3. To analyse patient and hospital performance data using Power BI and Excel.
4. To create interactive dashboards for healthcare insights.
5. To use AI tools such as Microsoft Copilot for report generation and analysis.
6. To improve healthcare decision-making through data-driven insights.

1.4 Scope of the Project

This project focuses on analysing hospital patient datasets collected from open-source platforms such as Kaggle.

The project uses Microsoft Excel, Power BI, and Microsoft Copilot for data cleaning, analysis, visualization, and AI-assisted reporting.

The project mainly focuses on healthcare analytics and dashboard creation. Real-time data integration and machine learning prediction models are not included.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

- Many healthcare organizations use business intelligence tools such as Power BI and Excel for healthcare reporting and operational analysis.
- Research studies show that AI tools and healthcare analytics help improve patient care, hospital efficiency, and decision-making processes.
- However, many existing systems lack simple and interactive dashboards suitable for non-technical healthcare administrators.

2.2 Tools and Technologies Review

- ☐ Microsoft Excel — used for data cleaning and preprocessing.
- ☐ Power BI — used for interactive healthcare dashboards.
- ☐ Microsoft Copilot — used for AI-assisted summaries and analysis.
- ☐ ChatGPT — used for documentation and content generation

2.3 Gaps Identified

- ☐ Existing systems are often complex for non-technical users.
- ☐ Limited use of AI-assisted analytics in student healthcare projects.
- ☐ Lack of interactive and simple healthcare dashboards

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

- ☐ Healthcare datasets were collected from Kaggle. The raw data was cleaned using Excel by removing duplicates, handling missing values, and correcting inconsistent formats.
- ☐ The cleaned data was analysed using Power BI to create interactive dashboards. AI tools such as Microsoft Copilot were used to generate summaries and insights

3.2 Modules / Components Used

- Module 1 — Data Collection
- Module 2 — Data Cleaning
- Module 3 — Data Analysis
- Module 4 — Dashboard Visualization
- Module 5 — AI Tool Integration

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

Raw Dataset → Healthcare Analytics System → Dashboard Output

DFD Level 1 — Detailed Process Flow

Breaks down the main process into sub-processes. For your project this may include:

Data Collection → Data Cleaning → Data Analysis → Visualization → Dashboard

3.4 Advantages

List the key advantages -

- ☐ Improves healthcare decision-making
- ☐ Provides interactive visual insights
- ☐ Reduces manual analysis effort

- ☐ Uses AI tools for faster reporting
- ☐ Easy to understand dashboards

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage.

4.1 Data Cleaning Results

The healthcare dataset obtained from Kaggle contained duplicate records, missing values, and formatting inconsistencies.

Data cleaning techniques such as duplicate removal, validation, and standardization were applied using Excel.

4.2 Visualization / Output Results

- ☐ Dashboard / Chart 1: [Describe what this visual shows — e.g., Monthly Revenue Trend Bar Chart]
- ☐ Dashboard / Chart 2: [Describe — e.g., Regional Sales Performance Map]
- ☐ Dashboard / Chart 3: [Describe — e.g., Top 10 Products by Profit Margin]

4.3 AI Tool Integration Results

Tool Used :

Microsoft Copilot and ChatGPT

Prompt Used :

- Summarize healthcare trends from dataset
- Identify operational inefficiencies
- Generate healthcare analytics report

Output Quality :

AI tools generated quick and useful summaries.

Enhancement :

AI reduced documentation time and improved analysis efficiency.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- ☐ Data cleaning improved accuracy.
- ☐ Dashboards helped visualize healthcare performance clearly.
- ☐ AI tools reduced report generation time.
- ☐ Data analytics improved healthcare insights.

5.2 Project Links

- ☐ GitHub Link of the Project: [Paste your GitHub repository link here]
- ☐ Video Recording of Project Demonstration: [Paste your YouTube / Google Drive / OneDrive video link here]
- ☐ Power BI / Dashboard Link (if applicable): [Paste published dashboard link here]

5.3 Limitations

- Dataset size was limited.
- No real-time healthcare data integration.
- Advanced AI prediction models were not included.

5.4 Future Work

- ☐ Real-time healthcare analytics
- ☐ Machine learning integration
- ☐ Mobile healthcare dashboards
- ☐ Cloud database integration

5.5 Conclusion

This project successfully demonstrates how data analytics and AI tools can improve healthcare decision-making through meaningful insights and visualization.

The integration of Power BI, Excel, and AI tools transformed raw healthcare data into actionable business intelligence and helped achieve all project objectives successfully.

REFERENCES

- Microsoft Excel Documentation
- Power BI Documentation
- Microsoft Copilot Documentation
- Edunet Foundation AICW Guidelines
- Kaggle Healthcare Dataset

APPENDICES

Appendix A — Sample Dataset

patient_id	age	gender	condition	procedure	cost	length_of_stay	admission	outcome	satisfaction
1	45	Female	Heart Disease	Angioplasty	15000	5	No	Recovered	4
2	60	Male	Diabetes	Insulin Therapy	2000	3	Yes	Stable	3
3	32	Female	Fractured	X-Ray and	500	1	No	Recovered	5
4	75	Male	Stroke	CT Scan and	10000	7	Yes	Stable	2
5	50	Female	Cancer	Surgery and	25000	10	No	Recovered	4
6	68	Male	Hypertension	Medication	1000	2	No	Stable	4
7	55	Female	Appendicitis	Appendectomy	8000	4	No	Recovered	3
8	40	Male	Fractured	Cast and Plaster	3000	6	No	Recovered	4
9	70	Female	Heart Attack	Cardiac Catheterization	18000	8	Yes	Stable	2
10	25	Male	Allergic Reaction	Epinephrine	100	1	No	Recovered	5
11	48	Female	Respiratory	Antibiotics	800	2	No	Stable	4
12	65	Male	Prostate Cancer	Radiation Therapy	20000	9	No	Recovered	3
13	30	Female	Childbirth	Delivery and	12000	3	No	Recovered	4
14	52	Male	Kidney Stones	Lithotripsy	6000	4	No	Recovered	4
15	58	Female	Osteoarthritis	Physical Therapy	4000	5	No	Stable	4
16	55	Female	Heart Disease	Angioplasty	15000	6	Yes	Recovered	3
17	62	Male	Diabetes	Insulin Therapy	2000	4	No	Stable	4
18	35	Female	Fractured	X-Ray and	500	2	Yes	Recovered	5
19	78	Male	Stroke	CT Scan and	10000	8	No	Stable	2
20	53	Female	Cancer	Surgery and	25000	11	Yes	Recovered	4

Appendix B — Screenshot Gallery

